# 5. Basic PL/SQL Questions

## 5.1 Basic PL/SQL Blocks

1. Write a PL/SQL block to display “Welcome to the Library”.
2. Accept a book ID and display that book’s title.
3. Accept a member ID and display the member’s name.
4. Display the total number of books in the BOOK table.
5. Display the total fine amount in the FINE table.
6. Display the titles of books published after 2020.

## 5.2 Cursor

1. Use a cursor to display all book titles.
2. Use a cursor to display all member names.
3. Use a cursor to display the title and price of each book.
4. Use a cursor to display members whose status is ACTIVE.
5. Use a cursor to display loans whose status is OVERDUE.
6. Use a cursor to display each book copy’s ID and availability status.

## 5.3 Procedures

1. Create a procedure that displays a book title when a book ID is given.
2. Create a procedure to add a new member to the MEMBER table.
3. Create a procedure to change a book’s price using its book ID.
4. Create a procedure to change a member’s status to SUSPENDED.
5. Create a procedure to display the number of books in a category.
6. Create a procedure to update the amount paid for a fine using its fine ID.

## 5.4 Functions

1. Create a function that returns a member’s name using the member ID.
2. Create a function that returns the total number of books.
3. Create a function that returns the number of members of a given member type.
4. Create a function that returns the number of AVAILABLE copies of a book.
5. Create a function that returns the total fine amount recorded for a given member ID.
6. Create a function that returns a staff member’s salary using the staff ID.

5.5 Triggers

1. Create a trigger to change a member's name to uppercase before inserting the member.
2. Create a trigger to stop a book's price from being set below zero.
3. Create a trigger to fill in the join date with today's date when it is not provided.
4. Create a trigger to set a book copy's status to AVAILABLE when a loan is marked RETURNED.
5. Create a trigger to stop the amount paid for a fine from being greater than the fine amount.
6. Create a simple trigger that displays a message when a new book is added.

5.6 Views

1. Create a view named BOOK_DETAILS showing each book's ID, title, category, publication year, and price.
2. Create a view named ACTIVE_MEMBERS showing members whose status is ACTIVE.
3. Create a view named AVAILABLE_BOOKS showing copies whose status is AVAILABLE.
4. Create a view showing book titles and publisher names, using the relevant tables if they exist in your database.
5. Create a view showing each book category and the number of books in it.
6. 48. Create a view showing staff names and their salaries.

## 4.5 subquery

1. Display the details of the staff member who has the lowest salary.
2. List the books published by a publisher located in a given city, for example, kochi
3. List the staff members who joined after the staff member with ID 502
4. List the staff members who have issued at least one loan.
5. Display the members who reserved the book Sapiens.
6. Display the publisher that published the book with the highest price.
7. List the books priced above the overall average book price.
8. List the members who belong to the same branch as member 1001, excluding member 1001
9. Find the second-highest staff salary.
10. Increase the salary by 5% for staff members who have issued more than two loans. Use a subquery in